

Table 3.3: Parameter values for EDM of titanium alloy

Parameter	Symbol	Unit	Level		
			1	2	3
Voltage	<i>Volt</i>	V	50	60	70
Peak Current	<i>Ip</i>	A	8	12	16
Pulse On Time	<i>Ton</i>	μs	6	8	10
Pulse Off Time	<i>Toff</i>	μs	7	9	11

The EDM parameters in Table 3.3 have direct impacts on the efficiency and quality of the EDM of the titanium alloy. The outcome is detailed in Table 3.4.

Table 3.4: *Ra* values from experimental dataset

Run	Volt (V)	<i>Ip</i> (A)	<i>Ton</i> (μs)	<i>Toff</i> (μs)	<i>Ra</i> (μm)
1	70	8	10	7	11.558
2	60	12	8	9	14.717
3	60	12	8	9	14.867
4	70	8	6	7	7.647
5	50	8	6	11	6.245
6	50	16	10	7	14.514
7	60	12	10	9	13.608
8	70	16	6	11	10.168
9	60	12	8	9	10.325
10	60	12	8	9	15.851
11	70	8	6	11	9.04
12	70	16	10	11	14.514
13	60	12	8	9	16.24
14	60	16	8	9	11.728
15	60	12	8	9	12.485
16	50	16	6	7	10.008
17	60	12	8	7	12.512
18	50	8	6	7	6.301
19	70	16	6	7	9.577

20	50	12	8	9	12.629
21	50	8	10	11	10.389
22	60	12	8	11	12.196
23	60	12	6	9	7.545
24	50	16	6	11	6.753
25	50	8	10	7	13.289
26	60	8	8	9	9.149
27	70	12	8	9	18.214
28	50	16	10	11	16.758
29	70	16	10	7	14.814
30	70	8	10	11	14.322